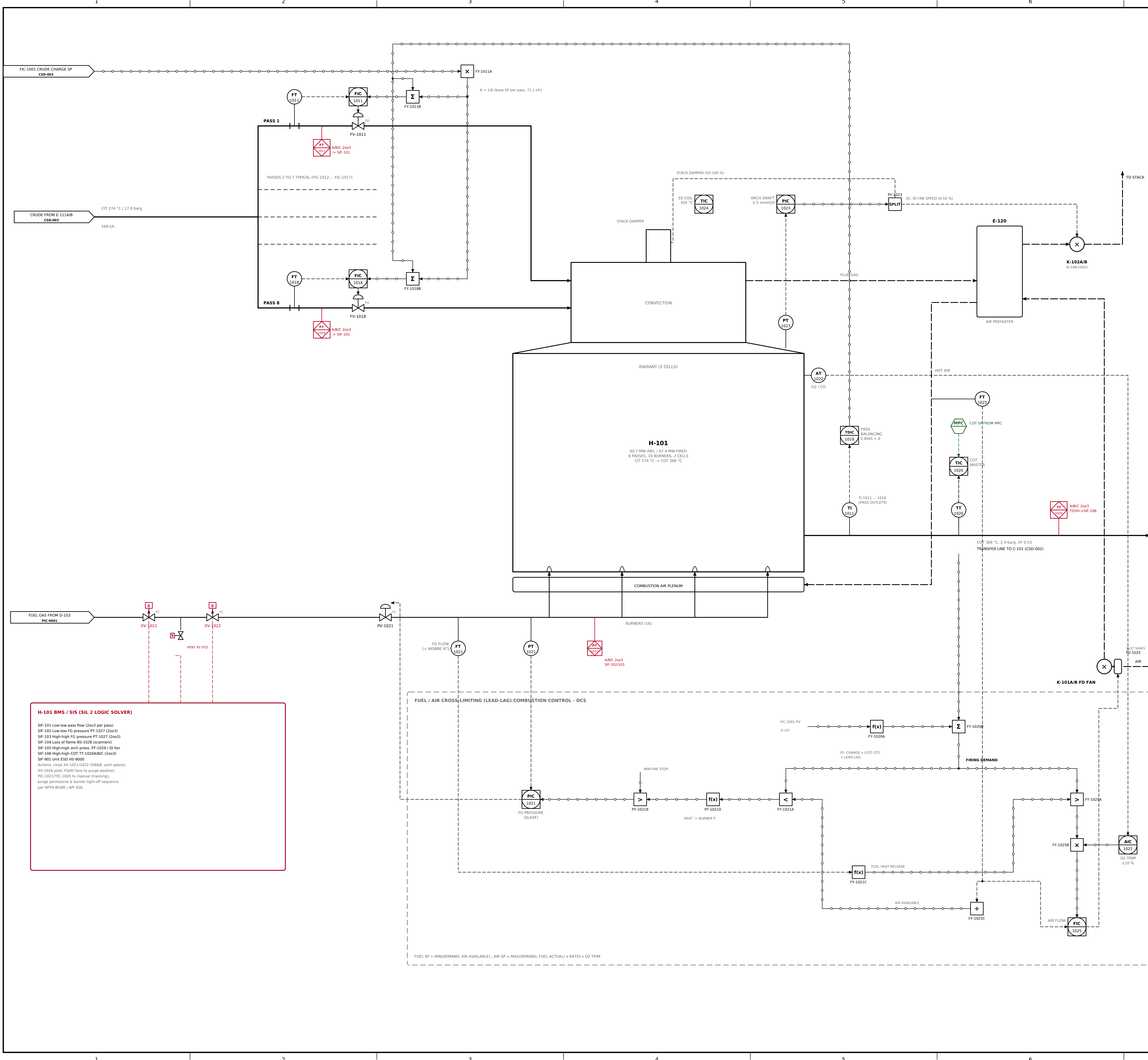




LEGEND (ISA-5.1 / ISA-5.4)

FIC

0000

DCS (BPCS) function, operator accessible

TC

0000

APC / MPC (L3 computer) function

Σ

Computing block: Σ sum/bias, $\times$ ratio/mult, $<$ low select, $>$ high select, $f(x)$ characteriser

$\rightarrow$

Electrical signal (4-20 mA / HART)

$\rightarrow$

Process line

SIS

0000

SIS / BMS function (SIL rated)

FT

0000

Field-mounted instrument

Control valve (FO/FC = fail open/closed)

$\rightarrow$

Software / data link (DCS internal)

$\rightarrow$

APC setpoint / MPC move

- CONTROL DESCRIPTION
1.

Crude charge master FIC-1001 (CSD-003) sets the base SP of the 8 pass flow controllers FIC-1011..1018 through FY-1011A (K = 1/8, 71.1 t/h per pass).
2.

Pass balancing TDC-1019 compares pass outlet temperatures TI-1011..1018 and computes biases FY-1011B..1018B whose sum is forced to zero, so total charge is unchanged; bias clamped $\pm 10\%$ of pass flow and frozen when any pass FIC is not in CAS.
3.

COT TIC-1020 (SP from MPC) adds to a feed-forward fix of charge x (COT-CTT) with lead lag (FY-1020A/B) to form the firing demand.
4.

Cross-limiting: fuel SP = low select of demand and air available (FY-1021A); air SP = high select of demand and actual fuel heat release (FY-1025A). On load increase air leads fuel; on decrease fuel leads air - no sub-stoichiometric firing.
5.

Fuel heat demand converted to burner pressure SP (burner curve, Wobbe-corrected, PY-1021A); min-fire stop PY-1021B; PIC-1021 manipulates PV-1021.
6.

O2 trim AIC-1022 (arch O2 with CO override) multiplies the air/fuel ratio, limited $\pm 10\%$. FIC-1025 positions the FD fan inlet vanes.
7.

Arch draft PIC-1023 (-2.5 mmH2O) split range: ID fan speed 0-50 %, stack damper 50-100 % (natural-draft fall-back on ID fan trip).
8.

All SIS/BMS functions are independent of the DCS (separate sensors, logic solver and SSOVs).

H-101 BMS / SIS (SIL 2 LOGIC SOLVER)

SIF-101

Low-low pass flow (Zoo3 per pass)

SIF-102

Low-low FG pressure PT-1027 (Zoo3)

SIF-103

High-high FG pressure PT-1027 (Zoo3)

SIF-104

Loss of flame BS-1028 (ccomers)

SIF-105

High-high arch press. PT-1029 / ID fan

SIF-106

High-high COT TT-1020A/B/C (Zoo3)

SIF-901

Unit ESD HS-9000

Actions:

close XV-1021/1022 (DB&B, vent opens),

XV-1026

pilot; FD/ID fans to purge position;

PIC-1021/TIC-1020

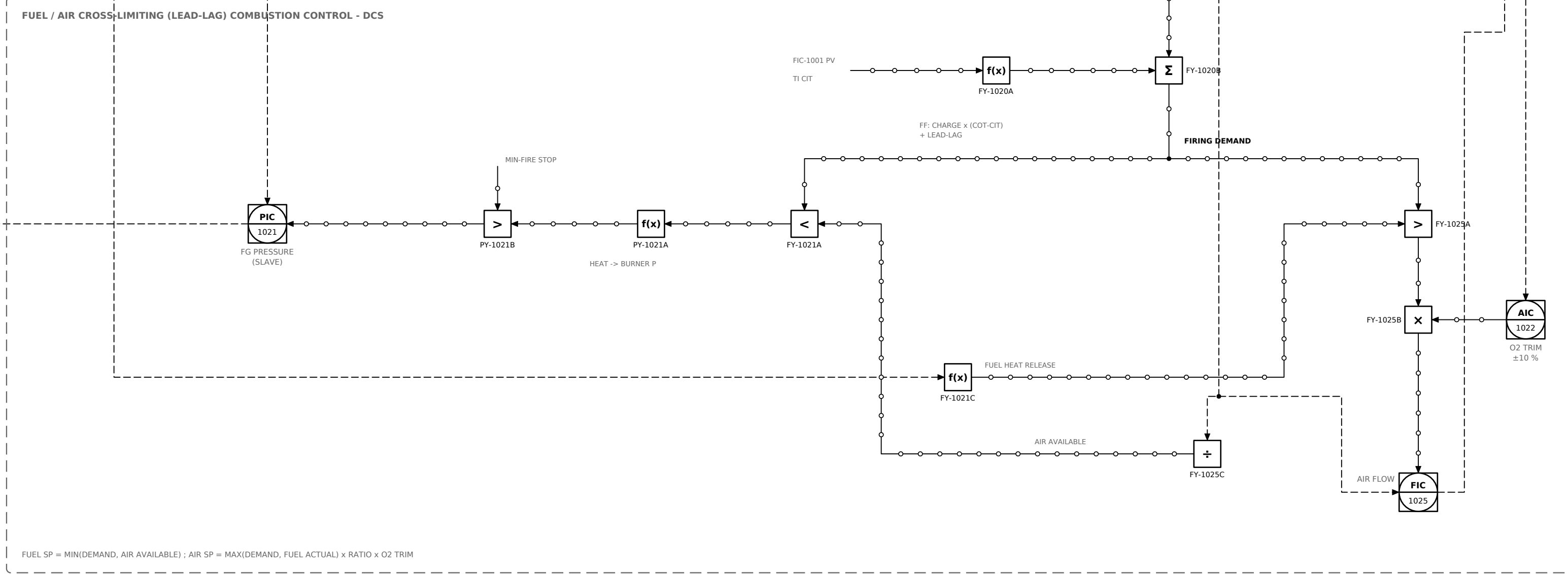
to manual (tracking);

purge

permissive & burner light-off

sequence

per NFPA 85/86 / API 556.



REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST		
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PROPOSED CONTROL SCHEME		
H-101 COMBUSTION, COT & PASS BALANCING CONTROL		
DISCIPLINE	STAGE	SCALE
INSTRUMENTATION	FEED	NTS
1 OF 6		
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DATE	2026-10-02	
DRAWING NUMBER	CFU-100-IC-CSD-001	
REV	A	